HARSH BRAHMBHATT

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PROFESSIONAL SUMMARY

AI Engineer focused on making LLM systems dependable in production. Built pipelines with component allowlists, output validation, regex-based auto-correction, and fallback handling so apps stay stable on bad model output. Deployed AI products on Vercel using Groq, Llama 3.3-70B, Gemma 3:1b and the Vercel AI SDK, plus a local-LLM data quality tool with Ollama, Pandas, and SQLite. Strong in Python, TypeScript, and REST API development.

EDUCATION

Vadodara Institute of Engineering (GTU) — B.E. in Computer Engineering 2023 – 2027

CGPA: 7.35

EXPERIENCE

Data Engineer Intern — NYSA Shipping Jul 3, 2026 – Jul 17, 2026 (2 weeks)

Ahmedabad, Gujarat (Online)

- Worked on data management, data validation, and report generation using the company's software systems (NYSA.IDEX.SOLUTIONS) to support day-to-day business processes.
- Supported software operations and ensured accuracy and consistency of operational data across reports and workflows.
- Recognized by the Director for sincerity, dedication, professionalism, and a strong willingness to learn.

Tech Stack: Python, SQL, Data Validation, Reporting

Intern — Atal Tinkering Labs, Shree Sagar Vidhyalaya (Mentor India Academy) January 2026 – April 2026

Atal Innovation Mission, NITI Aayog, New Delhi

- Completed an internship program at Atal Tinkering Labs under Mentor India Academy (MIA), an initiative of Atal Innovation Mission, NITI Aayog.
- Supported hands-on technology and innovation activities for school students, building practical problem-solving and teamwork skills.

Tech Stack: Language: C | Hardware: IoT instruments

PROJECTS

AI UI Generator — Text-to-UI with LLMs GitHub | Live Demo

- **Three-stage LLM pipeline:** a Planner breaks the prompt into a UI structure, a Generator writes React code from that plan, and an Explainer describes the design choices in plain language.
- **Model/API integration:** integrated Llama 3.3-70B through the Groq API using the Vercel AI SDK, with stage-specific prompts and streamed responses.
- **Iterative generation:** follow-up prompts modify the existing UI instead of regenerating it, preserving prior work; supported by diff view, undo/redo, and version restore.
- **Validation layer:** component allowlist, regex-based auto-correction of malformed output, and JSX validation before rendering.
- **Error handling:** invalid generations fall back to a safe error card, so the app never crashes on bad model output.
- Deployed on Vercel with a split-pane workspace: live preview with responsive device toggle, code editor, and export to .tsx or standalone HTML.

Tech Stack: Next.js 16, React 19, TypeScript, Tailwind CSS v4, Vercel AI SDK, Groq (Llama 3.3-70B), React-Live, Vercel

NeuralLab — Interactive Artificial Neural Network (ANN) Learning & Debugging Platform GitHub | Live Demo

- Developed an interactive web-based platform featuring an Artificial Neural Network (ANN) implemented from scratch in TypeScript without external machine learning libraries.
- Implemented forward propagation, backpropagation, loss calculation, gradient computation, and neural network training with configurable architectures and activation functions.
- Built real-time visualizations of network layers, neurons, weights, biases, gradients, training loss, accuracy, and decision boundaries.
- Integrated optimization algorithms, including SGD, Momentum, and Adam, with an optimizer comparison arena to analyze training performance.
- Implemented gradient checking and training diagnostics to validate mathematical correctness and identify potential training issues.
- Added interactive datasets, guided learning missions, experiment history, and JSON report export for hands-on ANN experimentation.

Tech Stack: Next.js, React.js, TypeScript, Tailwind CSS v4, Recharts, Artificial Neural Networks (ANN), Forward Propagation, Backpropagation, Gradient Descent, SGD, Momentum, Adam Optimizer, Gradient Checking.

AI-Powered ETL Pipeline | Datasense AI GitHub | Live Demo

- Built an interactive Streamlit dashboard to upload, preview, profile, and clean CSV datasets using 4 automated rules: duplicate removal, whitespace trimming, mean imputation, and “Unknown” fill.
- Developed a data quality scoring engine across 4 dimensions: completeness, uniqueness, validity, and consistency, generating actionable data-cleaning recommendations.
- Implemented automated data profiling to identify missing values, duplicate records, data types, and statistical summaries for improved data quality assessment.
- Integrated an AI-powered chat interface using Ollama Gemma 3:1b to answer dataset-related questions and generate insights, explanations, and data-driven recommendations.
- Persisted cleaned datasets in SQLite and enabled CSV export for further analysis, supporting an end-to-end data processing workflow.

  Tech Stack: Python, Pandas, Streamlit, SQLite, Ollama (Gemma 3:1b)

SpendSense — Personal Finance Management App (Full-Stack) GitHub | Live Demo

- Built a full-stack finance app with 5 core modules (accounts, transactions, budgets, savings goals, recurring payments), deployed live on Vercel.
- Developed interactive spending analytics dashboards, an AI-powered finance assistant (Groq API), and secure authentication using JWT and bcrypt.

  Tech Stack: React, TypeScript, Tailwind CSS, Node.js, Express.js, MongoDB, Mongoose, Recharts, Groq API, JWT, bcrypt, Vercel

TECHNICAL SKILLS

AI / ML: LLM Integration (Groq, Llama 3.3-70B, Ollama), Vercel AI SDK, Prompt Engineering, Pandas, NumPy, Artificial Neural Networks (ANN), Forward Propagation, Backpropagation, Gradient Descent, SGD, Momentum, Adam Optimizer, Gradient Checking
Languages: Python, JavaScript, TypeScript, C, SQL
Frontend: React.js, Next.js, TypeScript, JavaScript (ES6+), HTML5, CSS3, Tailwind CSS, Recharts, Responsive Web Design
Backend: Node.js, Express.js, REST APIs, API Development, Authentication & Authorization (JWT, bcrypt)
Database: MongoDB, Mongoose, SQLite, MySQL
Cloud & Tools: Git, GitHub, Vercel (Deployment), Streamlit
Concepts: Full-Stack Development, ETL & Data Pipelines, Data Cleaning, Data Structures & Algorithms, OOP, DBMS, SDLC

ACHIEVEMENT

- **1st Rank in Robotics Competition** at Vadodara Institute of Engineering — designed and built a line-following robot car and won first place at the college level.

  Tech Stack: Language: C | Tool: VS Code | Hardware: Arduino IDE, IR Sensor Array, Motor Driver (L298N)